Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/02/2012
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Project Title:	Development and use of human 3-Dimensional tissue culture models for the study of space radiation effects on the central nervous system		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032-3702	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Space Radiobiology NNJ09ZSA001N
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: received extension to 10/31/2011 (from 10/31/2010) per NSSC (01/2011)		
Key Personnel Changes/Previous PI:			
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Task Description:	Human space exploration involves exposure to types of radiation not normally encountered on earth. As well as carcinogenesis, the effect of these particles on the human central nervous system is of particular concern since impairment of cognitive function could not only compromise missions but cause later effects on the quality of life. Studies in animals have shown that space radiation particles can have adverse effects on cognition and behavior. The effect on neurogenesis has also been documented and proposed to be a cause of cognitive loss. Information regarding the effects of space radiation on the CNS is however, still in its infancy due to the difficulty of extrapolating data from animals and cell lines to the intact human being. The primary objective of the research in this proposal was to increase our understanding of the effects of space radiation on the human central nervous system with an emphasis on neurodegeneration. We approached this question by using available human stem cells to create 3-dimensional tissue models. These can represent human neural tissue more closely than 2-dimensional monolayers. Human cells of three of the main types in the brain were cultured and differentiated in 3-D matrix gels with a view to determining the effects of space radiation by assaying for specific endpoints related to degeneration in neurons. This could be achieved by live and fixed stain 3-D imaging together with assays for secreted proteins. 3-D cultures are grown in gels as monocultures and as a multi-cell 'neurovascular unit' built on an existing model of human vessels. Such a model can complement animal and cell studies to produce new and more complete data on the effects of space radiation on the human nervous system. This award was originally for one year only but a no-cost extension was acquired for a second year. During the first year we developed the tissue models to an extent that they could be used in further degeneration studies. During the second year on a no cost extension we applied the 3-D tissue models along with other monolayer models to develop assays for a projected study on the effects of space radiation on neurodegeneration with specific emphasis on the Blood-Brain Barrier (BBB) and angiogenesis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of 3-Dimensional human tissue models from normal human cells and stem cells is a relatively new technology but one with great potential in many fields of medical research. Tissue models can more accurately depict human tissue since the cells can be arranged spatially as they would be in vivo and can interact with each other as they would in the human body. A neurovascular unit can be used for basic research on many aspects of the human brain. These include regeneration, synaptic function, and degeneration. Because the tissue model is derived from individual cells, each cell type can be altered genetically before it is incorporated into the model. Such modification has a great advantage in that specific aspects of brain function can be targeted.
Task Progress:	This report is divided into two parts. The first part describes our work on developing the 3-D tissue models that represent the human neurovascular unit. The second part describes our work on developing assays for future investigations targeted to the function of the Blood Brain Barrier and angiogenesis. The 3-D tissue models are incorporated into a subset of these assays. Tissue model development We have attempted to approach the development of the human brain tissue model in a methodical and structured way. First, we acquired at least 3 human cell types for each of the components of the model; neurons, astrocytes and endothelial cells, to ensure the best possible model and provide back-up cells. Second, stem cells are the most suitable source of neurons since they are closest to normal, so we have developed our own multistep protocol for the generation and differentiation of normal human neurons from these cells and successfully detected expression of neuronal markers and degeneration markers in 3D monocultures. Astrocytes and neuroblastoma are differentiated by treatment with retinoic acid and endothelial cells can be readily induced to form microcapillaries in 3-D matrices. The strategy for constructing tissue models is to embed neurons and astrocytes into protein gel matrices after they have been differentiated into the respective cell type then induce morphological or 'endstage' differentiation while they are in the matrix. Endothelial cells can then be added later, given the right factors, and allowed to grow into the matrix and form blood vessels as in vivo during angiogenesis. We have tested different media types and different pairings of cell types and found that the requirement of endothelial cells is not entirely compatible with that of neurons and astrocytes. Endothelial cells, which line the vessels and are in contact with blood in vivo require serum for vessel formation. Neurons and Astrocytes, normally behind the blood brain barrier (BBB), require serum-free conditions to differentiate in matrices. However, we were able to create a prototype neurovascular unit using the above protocol by reducing the serum concentration and relying on trophic support from astrocytes for the maintenance of vessels. We are currently improving the longevity of these models and substituting neurons from stem cells for the neuroblastoma used in our prototype model. At this stage we assessed that tissue models were developed enough to be used in certain assays for the investigation of the effects of space radiation on the brain. Model development could continue for much longer but application at this stage is appropriate for advancement of a project in the Central Nervous System section of the human research program. Assay development The remainder of this award was spent developing assays for the application of a new grant award proposing to study the effects of space radiation on the BBB and angiogenesis. The tissue models we developed in the first part of this award can be incorporated into the assays that assess tight junctions and provide an additional relevant endpoint. Vessels are central players in neurodegeneration including Alzheimer's and the assay of their function can give much information about the effects of both harmful and beneficial agents. Angiogenesis and the BBB are highly important in maintaining vessel function. Their biology and pathology are intertwined, each can affect the other in normal and pathological conditions. The assay development described here is directed towards a study to determine the effects of space radiation on the BBB. For BBB studies we have developed 4 assays to determine the function of the endothelial barrier. First, transendothelial resistance measurement is achieved by using equipment that measures electric cell-substrate impedance sensing (ECIS). Resistance monitoring of monolayers is easy and continuous. This assay is useful for determining parameters such as the time course of the effects of radiation. This information can then be utilized to design experiments that measure the functioning of the BBB more directly. These include, a second assay using tissue inserts to determine the rate of labeled molecules permeation through the cell layer on the insert. This would be a direct measurement of the passage of molecules of varying sizes through the endothelial cell layer. This assay does not utilize the neurovascular unit but can

	still include a cell mixture of endothelial cells and astrocytes or pericytes. Endothelial cells are plated on one side of the membrane and the other cell types are plated on the other side of the membrane. Thus it will represent the human blood-brain-barrier more closely than endothelial cells alone. A further 2 assays for localization of the junction proteins that seal the BBB using immunolabeling of these molecules. First examining the junction proteins in normal monolayers and second examining the junction proteins in our 3-D human tissue models. In the latter case we can also include other cell types that grow with vessels in humans. Further assays include endpoints more directly applied to human neurodegenerative diseases. One such endpoint is the metabolization and localization of the protein involved in Alzheimer's disease – the amyloid precursor protein (APP). Immunolabeling has enabled us to detect this protein, inside the cell and at the cell surface. Furthermore, we can also detect, the cleaved products of APP that are released into the media, thus we are able to track the expression, localization and cleavage of this degeneration marker after irradiation and can therefore monitor some of the events that occur during Alzheimer's degeneration. Neuroblastoma cells, which can be easily differentiated, have a particular advantage in that they over express APP. To conclude, we have achieved the primary goal of this short award - human 3-D tissue models developed sufficiently for use in novel assays to determine degenerative markers. We have also used this award to develop other assays for use in future studies that address a pivotal part of the brain involved heavily in many neurodegenerative disorders – the brain blood capillaries.
Bibliography Type:	Description: (Last Updated: 03/04/2024)
Articles in Peer-reviewed Journals	Grabham P, Hu B, Sharma P, Geard C. "Effects of ionizing radiation on three-dimensional human vessel models: differential effects according to radiation quality and cellular development." Radiat Res. 2011 Jan;175(1):21-8. Epub 2010 Nov 4. PMID: 21175343 , Jan-2011
Articles in Peer-reviewed Journals	Hu B, Grabham P, Nie J, Balajee AS, Zhou H, Hei TK, Geard CR. "Intrachromosomal changes and genomic instability in site-specific microbeam-irradiated and bystander human-hamster hybrid cells." Radiat Res. 2012 Jan;177(1):25-34. Epub 2011 Nov 11PMID: 22077336 , Jan-2012
Articles in Peer-reviewed Journals	Grabham P, Bigelow A, Geard C. "DNA damage foci formation and decline in two-dimensional monolayers and in three-dimensional human vessel models: Differential effects according to radiation quality." International Journal of Radiation Biology. 2012 Jun;88(6):493-500. Epub 2012 Apr 30. PMID: 22449005 , Jun-2012